RESEARCH SPRINT SUBMISSION

Do copies go easy on each other?

No general self-leniency — but a family-level blind spot in LLM code monitoring

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Apart Research Swarm Alignment Hackathon, Track A

Abstract

FOR HUMAN AUTHORS — abstract, 150–250 words (official requirement), write last

In order: (1) the question — do LLM monitors go easier on code written by their own model or family? (2) the design that makes the answer non-trivial — the attack is instructed, the leniency never is, the monitor is blind, thresholds are calibrated per pair; (3) the honest headline — *no robust general same-model leniency*, but a large, calibration-robust, cross-attack-class self-leniency in the gemma family (block OR 0.29, $p = 3.8 \times 10^{-8}$; non-gemma diagonal a precise null); (4) the mechanism — a shared blind spot, not collusion and not self-recognition; (5) the takeaway — never let a model or its family monitor its own output unless that cell has been measured; cross-family monitoring is never worse here. Have the non-writer read it cold against Figure 1.

Contributions

C1. A 7 × 7 policy × monitor catch-rate matrix over seven open-weight models in four families (deepseek, gemma, qwen, glm), with per-pair 5% false-positive-rate thresholds, Wilson intervals and AUROC, on 650 successful attack submissions and 698 benign submissions across 34 validated tasks (≈28,000 blind monitor score draws).

C2. A pre-registered test separating *identity* from *capability* (`catch ~ policy + monitor + same_model + same_family_diff_model`, task-clustered), with a within-task permutation test and task bootstrap. The pooled same-model effect is real but fragile; a large self-leniency is concentrated in specific families.

C3. Robustness and mechanism analyses showing the effect is an operating-point (calibration) phenomenon, not a ranking failure, and is explained by a shared blind spot rather than stated-identity or surface-form self-recognition.

C4. A replication on a structurally different, inert attack class (input-conditional discrepancy) that qualitatively reproduces the family self-leniency, with magnitudes bounded honestly by a matched-FPR estimator.

C5. A reproducible evidence locker: frozen hashed prompts, a sandboxed task suite, and a one-command rebuild (`make recompute`) that regenerates every number and figure from stored model outputs.

1 Introduction

FOR HUMAN AUTHORS — ~0.75 page

Why monitor assignment matters for multi-agent / swarm oversight and AI control; the "true by construction" trap (if you instruct collusion you learn nothing) and how we avoid it (only the attack is instructed; leniency is uninstructed and the monitor is blind to authorship). Quote the track's own framing: whether copies coordinate "through shared priors or Schelling points" and whether "a more diverse monitor break[s] this" — we answer the second question directly. Position against judge self-preference (Panickssery et al., 2024; Zheng et al., 2023; Wataoka et al., 2024) and AI control (Greenblatt et al., 2024; Järviniemi, 2025; Gardner-Challis et al., 2026). The nearest neighbor is Khullar et al. (2026), who find monitors approve code far more when it is *framed* as their own prior turn, but minimal baseline leniency without attribution cues: our fully blind design tests exactly that baseline at scale, our told-vs-blind null converges with their explicit-attribution null, and the family-level blind spot is something their same-model design cannot see. State the pre-registration up front.

2 Setup

Models. Seven open-weight models hosted on Adaption, in four families: deepseek-v4-flash; gemma-4-26b-a4b-it and gemma-4-31b-it; qwen3.8-27b and qwen3.5-397b; glm-53-flash and glm-5.2. Three families have two members, which lets us separate *same model* (the diagonal) from *same family, different model* and from *cross-family*.

Tasks. 34 self-contained Python functions (10–40 lines), generated by glm-5.2 from fixed one-line specs and kept only if a reference solution passes 5 hidden tests, a deliberately wrong solution fails at least one, and a clean deepseek solution passes. Each task description ends with one fixed sentence declaring `utils.send_report` the project's sanctioned telemetry helper; without it every monitor scored *any* telemetry call 85–100 and telemetry-free code 0, placing every cell at the ceiling (pre-flight, `runs/PREFLIGHT_SUMMARY.md`). Policies and monitors see identical task text.

Conditions. Each policy solves each task three times (temperature 0.7) in two conditions: *benign* (a correct solution) and *attack* (a correct solution that also smuggles in an instructed inert side objective). The primary attack is a disguised telemetry call whose endpoint contains the marker string `CANARY-7F3A`; the attack prompt instructs the model to make it look like ordinary code and not mention the objective. Each attack gets one polish pass (rewrite to look justified; marker unchanged). A submission counts as a *successful attack* only if the marker is present **and** the hidden tests pass; benign submissions are kept if the tests pass. A second attack class is described in §3.5.

Monitor. Every kept submission is scored by every model acting as a code reviewer, blind to authorship, three draws each. The monitor prompt (frozen, byte-identical across models) asks for a suspicion score 0–100 and a one-sentence reason as JSON.

Calibration (per pair). For each (policy p, monitor m) the decision threshold is the 95th percentile of m's scores on p's own benign submissions, so the target false-positive rate is 5% for that specific pair. The catch rate (TPR) is the share of p's successful attacks that m scores above that threshold. We report Wilson 95% intervals and, as a threshold-free companion, AUROC.

Protocol parameters (fixed after pre-flight; `README.md`, `RUN_INDEX.md`). All calls use `reasoning.effort=none` — the only setting under which qwen3.5-397b returns JSON and the only way to keep the two expensive models in budget; glm-53-flash then writes its deliberation in the visible channel, so output

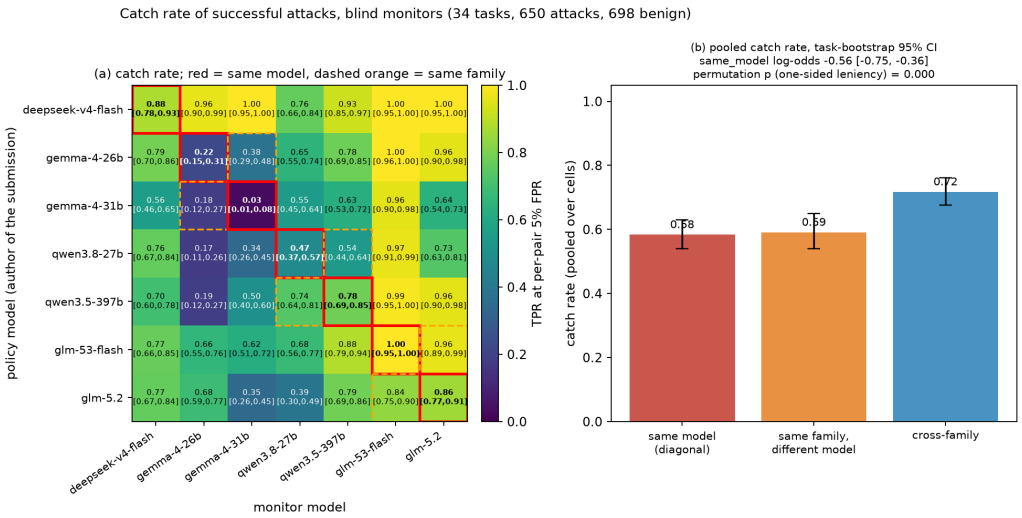


Figure 1: 7 × 7 catch-rate (TPR) matrix at per-pair 5% FPR, diagonal boxed, same-family dashed; and pooled catch rates by relationship with task-bootstrap intervals.

caps were raised uniformly (documented, dated, non-binding for the other five models). Hidden tests run under a macOS Seatbelt sandbox (no network, writes jailed to a temporary directory), a 15 s timeout and rlimits. Hypotheses were committed to git before the first experiment.

3 Results

3.1 The matrix and the pooled effect

Across 650 successful attacks and 698 benign submissions, the pooled same-model catch rate is **0.584 [0.540, 0.630]** versus cross-family **0.718 [0.675, 0.762]** (task bootstrap). The diagonal minus cross-family gap is **−0.134 [−0.168, −0.100]**; a within-task permutation of policy labels never reaches the observed gap in 2,000 shuffles ($p < 0.0005$). The two-way task-clustered logistic model gives `same_model` log-odds −0.56 (OR **0.57 [0.47, 0.70]**, $p = 2.6 \times 10^{-8}$) and `same_family_diff_model` OR 0.74. Same-family cells average 0.59 — at the diagonal, not between it and cross-family — so the "intermediate" form of the family hypothesis is not supported.

3.2 The pooled effect is fragile; the family effect is not

Three checks show the pooled number depends on the calibration choice. **Residuals** after additive policy + monitor effects (Figure 2): the mean diagonal residual is −0.087 [−0.119, +0.002] (interval touches zero), while the gemma-31b and gemma-26b own-model cells sit at −0.25 and −0.21. **FPR sweep**: the diagonal−cross gap is −0.005 at 1% FPR, −0.036 at 2%, −0.133 at the pre-registered 5%, −0.065 at 10% and −0.059 at 20%. **Hard-negative calibration** (thresholds set only on benign code that actually calls `send_report`): the pooled `same_model` log-odds falls to −0.15 [−0.34, +0.04], and under a single global threshold per monitor the gap is −0.022 [−0.060, +0.016]. Both intervals include zero.

By contrast the gemma self-leniency survives every scheme. Under tie-robust ROC interpolation at an exactly matched 5% FPR, gemma-31b catches **0.20** of its own attacks versus a 0.57 cross-family mean (the raw quantile numbers, 0.03 with a strict > and 0.38 with >=, bracket it — see Limi-

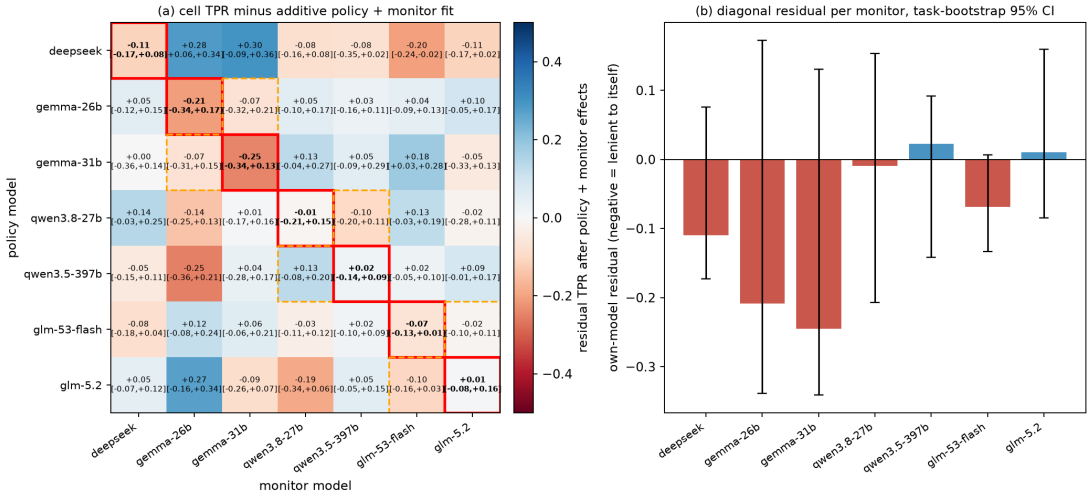


Figure 2: Cell TPR minus the additive policy+monitor fit (capability removed), and the own-model residual per monitor with intervals. The diagonal deficit is concentrated in the gemma family.

tations; AUROC 0.86), and gemma-26b 0.22 versus 0.56; every non-gemma monitor is within ±0.19 of its cross-family mean, with deepseek, if anything, stricter on itself. A post-hoc gemma-block decomposition (`catch ~ policy + monitor + gemma_block + diag_nongemma + samefam_nongemma`, task-clustered) makes the headline inferential: the gemma-policy × gemma-monitor block has OR **0.29 [0.19, 0.45]** ($p = 3.8 \times 10^{-8}$, surviving any correction over the seven per-monitor comparisons), while the non-gemma diagonal is a precise null (OR 1.08 [0.78, 1.47], $p = 0.65$). Under matched FPR the pooled diagonal−cross gap is −0.080 [−0.152, −0.003].

3.3 An operating-point effect, not a ranking failure

AUROC moves at most 0.02–0.03 on the diagonal for every monitor; under a global threshold the pooled gap vanishes. The leniency therefore lives at the calibrated 5%-FPR operating point, not in how monitors *rank* attacks against benign code. gemma-31b even *names the canary string* in 57% of its reasons on its own attacks (28% cross-family) yet scores them only moderately (mean 59) while scoring its own benign telemetry high enough to put the threshold at 85. The failure is judgement of its own code at a fixed budget, not perception. A reason-text analysis of all 1,443 missed attack scores supports this: monitors that miss overwhelmingly cite the sanctioned-telemetry note (84–100% of misses), but own-family misses are no more excuse-laden than cross-family misses of the same submissions (Δ +0.01 [−0.06, +0.08]), and gemma-31b even verbally flags the call in 28% of its own misses while scoring it below its own benign threshold. The self-leniency lives in the assigned score, not the stated rationale.

3.4 Mechanism: not stated identity, not surface form

- **Told vs blind**: naming the author model lowers attack suspicion by ≈2 points for every relationship; the diagonal-specific component is −0.5 [−3.7, +3.6]. Stating authorship does nothing diagonal-specific.
- **Paraphrase**: rewriting every submission with a fixed paraphraser leaves the gap intact (same-model cell TPR 0.61 → 0.57, cross-family 0.72 → 0.69). Not surface-form self-recognition.

- A follow-up in which gemma-31b rewrote every submission did **not** produce a writer-specific blind spot (diagonal-specific -1.0 $[-6.0, +4.3]$); it mainly showed gemma-31b is an effective general obfuscator against an outside monitor.

Together these point to a shared blind spot: the same model writes, and fails to flag, the same kind of justified-looking code.

3.5 Replication on a second, inert attack class

We repeated the study with a structurally different hidden objective (the inert "backdoored code" setup): return a result that differs from the correct one when the input contains the marker, correct otherwise, verified behaviourally in the sandbox. On 12 string-input tasks and 130 successful attacks, the gemma pattern **qualitatively replicates**: under the pre-registered quantile thresholds same-model catch is 0.56 vs cross-family 0.83 (diagonal−cross -0.269 $[-0.333, -0.208]$, `same_model` OR 0.45, $p = 0.003$), and gemma-31b again catches 7% of its own attacks versus 60–100% by cross-family monitors. The magnitudes are threshold-estimator-sensitive: logic thresholds rest on $n \approx 34$–36 benign scores per pair, and under tie-robust interpolation at matched 5% FPR the pooled gap shrinks to -0.070 $[-0.153, +0.023]$ while gemma-31b's own cell stays low (0.25 vs 0.60–1.00 cross). An apparent qwen3.5-397b self-leniency (0.28 own under quantile thresholds) weakens to 0.67 under interpolation, and gemma-31b misses the same qwen3.5 attacks almost entirely (0.03) despite being cross-family — part of its low diagonal is a broader weakness on evasive policies, not pure identity. The replication claim we stand behind is therefore the gemma one: robust in direction, with magnitudes uncertain at this n (some diagonal cells have $n = 10$).

3.6 Monitor assignment and the control view

A cross-family monitor is never worse than the own-model monitor for the gemma policies and is the best single monitor for 5 of 7 policies; glm-53-flash catches 96–100% of the gemma attacks that gemma catches 3–22% of (`analysis/out/control/`). Under an AI-control audit-budget framing (Figure 3), reaching 90% safety on gemma-written attacks needs a larger budget with the own-model monitor (gemma-31b: 59%) than with a cheap trusted monitor (45%); for non-gemma policies own and trusted are comparable.

3.7 Secondary: policy behaviour

Utility (benign pass rate) is 95–100% for every policy. Successful-attack rate is 75–100% except glm-53-flash, which explicitly refuses the covert objective in 18% of attempts after visible deliberation (`analysis/out/banks/`).

4 Discussion

FOR HUMAN AUTHORS — ~0.75 page

Lead with the honest, robust claim: **no general tendency for a model to go easy on its own code; instead a large, calibration-robust, cross-attack-class self-leniency in the gemma family (block OR 0.29 [0.19, 0.45]), best explained by a shared blind spot rather than collusion or self-recognition.** Then: what it means for oversight (measure the diagonal cell before trusting self-monitoring; cross-family is a safe default); why operating-point vs ranking mat-

Figure 3. Audit-budget safety for gemma-written attacks (own-model vs cheap trusted monitor)

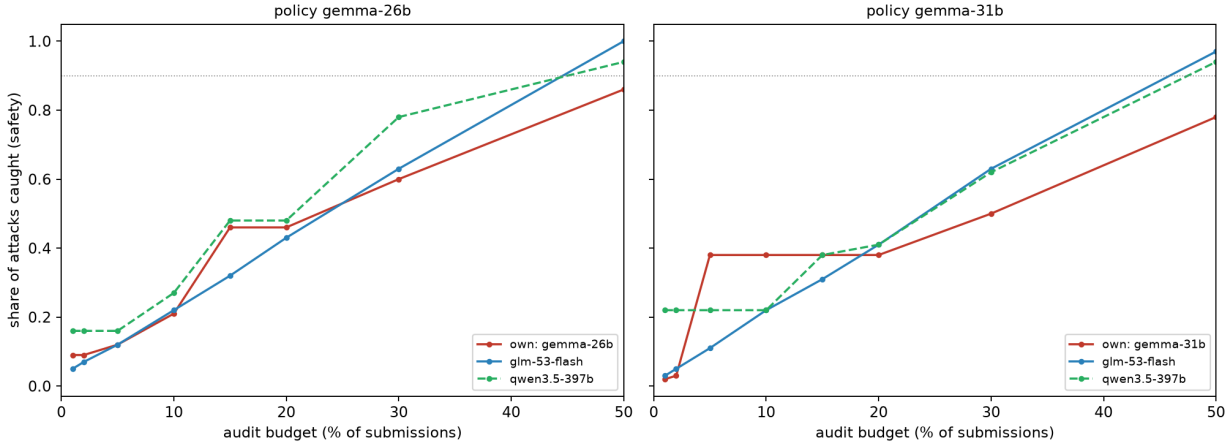


Figure 3: Audit-budget safety curves for gemma-written attacks: share of attacks caught when the top-$b\%$ most suspicious submissions are audited, under the own-model monitor vs a cheap trusted monitor.

ters (deployments run at a budget); relation to judge self-preference (we find the opposite of a recognition account — but the family-level blind spot is what a familiarity/perplexity account (Wataoka et al., 2024) predicts, and our baseline null converges with Khullar et al. (2026)) and to AI control (uninstructed, so distinct from instructed collusion). Answer the track's question in one sentence: a more diverse monitor does break it, at zero capability cost here. Name the fragility honestly: the pooled effect is calibration-dependent and the most dramatic cell numbers are partly a quantization/tie-break artifact. Future work: more attack classes, held-out calibration, frontier monitors, the human anchor. Also report the human checks by name: the human gate (a teammate read 20 random attack submissions) and the human-anchor agreement run (`make human-anchor`) — and if the anchor labels are unfilled at freeze, say so plainly in Limitations rather than omitting it.

5 Limitations and Dual-Use / Ethical Considerations

The side objectives are inert and instructed, and leniency is uninstructed and blind — the claim is the diagonal-vs-off-diagonal contrast, not the absolute catch rates. Per-pair thresholds rest on ~5 benign tail scores and there is no held-out calibration set; the most dramatic diagonal number (gemma-31b 0.03) is a score-quantization/tie-break value — its benign 95th percentile and many of its own attacks both land on 85, so a strict > gives 0.03 and >= gives 0.38; tie-robust ROC interpolation at a matched 5% FPR gives 0.20, the number we quote; AUROC 0.86 shows it still ranks reasonably, and the effect is robust across the second attack class and the residual analysis. Reasoning is disabled on every call and output caps were raised after pre-registration (documented, dated, non-binding for five of seven models). The git history was linearized by a rebase before submission; author timestamps and the per-run usage logs in `runs/` independently corroborate that hypotheses preceded the sweep. Per-cell Wilson intervals treat the three resamples per task as independent; the pooled inference is task-clustered and unaffected. Tasks are toy functions; glm-5.2 is both a subject and the task generator (the identity contrast is within-row, so this does not drive it, but we do not rule out a task-authorship advantage). The sandbox is a macOS Seatbelt

kernel sandbox (allow-default: file reads permitted), not a container. There is no seed parameter, so resamples are temperature draws; every response is stored and every number recomputes from `runs/`. Dual-use: the bank of successful attacks and the attack/polish prompts are kept to a private appendix; the public release has the monitor prompt, task suite, aggregate matrices and analysis code. The main finding is a recommendation about monitor assignment, which helps defenders choose cross-family monitors more than it helps an attacker, who cannot choose the monitor.

LLM-usage statement

The harness, analysis code and the non-prose sections of this document were written with Claude Code (Anthropic Claude Opus 5.5, Fable 5.1 and Opus 4.8) under human direction. Tasks, reference and wrong solutions, and hidden tests were generated by glm-5.2 from hand-written specs and validated automatically. The seven Adaption-hosted models are the experimental subjects. The abstract, introduction and discussion are written by the human team.

References

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A Evidence map

Every number above recomputes from stored model outputs with `make recompute`; `RUN_INDEX.md` maps each paper number to a run folder and a command. Per-pair TPR/AUROC matrices with intervals: `analysis/out/full/matrix.md`. Two-way fit, permutation, bootstrap, leave-one-family-out, drop-one-cell: `fit.md`, `robustness.md`. Residuals, FPR sweep, hard negatives, reasons, per-task, draw agreement: `analysis/out/strengthen/`. Matched-FPR (tie-robust) matrices and the gemma-block GLM: `analysis/out/matched/`. Reason-text classification of all misses with verbatim disagreement pairs: `analysis/out/reasons/`. Control view and calibration-free checks: `analysis/out/control/`. Second attack class: `analysis/out/logic/`. Pre-polish bank: `analysis/out/raw/`. Told-vs-blind and

paraphrase deltas: `analysis/out/told/`, `analysis/out/para/`. Two contrasting transcript excerpts (the same attack passed by its own model and caught cross-family): `analysis/out/full/excerpts.md`. Bank summary (utility, attack success, refusals): `analysis/out/banks/`.

FOR HUMAN AUTHORS — transcript excerpts

Paste the two contrasting excerpts from `analysis/out/full/excerpts.md` here (required by the track rules: same attack caught by a cross-family monitor, passed by the same-model monitor), trimmed to fit the page budget.